

Snowbasin Congestion Engine

CS-6580 Final Presentation

Team Wildcats

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Live App: <https://dashboard-snowbasin-wildcats.vercel.app/>

1) The Hook: Why This Matters

Stakeholder

Snowbasin Resort skiers/snowboarders managing better arrival times

Business Problem

Traffic surges on Trappers Loop (SR-167) create:

- arrival delays,
- parking bottlenecks,
- staffing misalignment,
- poor guest experience.

Project Goal

Forecast hourly traffic demand **up to 72 hours ahead** so users can act before congestion forms.

2) CRISP-DM Journey (Semester Arc)

- **Business Understanding:** define congestion risk + operational decisions.
- **Data Understanding:** collect traffic, weather, holiday context.
- **Data Prep:** normalize multi-year, mixed-format sources.
- **Modeling:** benchmark baseline vs advanced approaches.
- **Evaluation:** compare metrics + user utility.
- **Deployment/Review:** production-style dashboard + user-tested workflow.

Sprint 6 focus: lock code, polish docs, present decision-ready insights.

3) Data Pipeline (High-Level)

Inputs

1. **Traffic counts** (Trappers Loop, multi-year hourly lane data)
2. **Weather observations** (SBBWK station)
3. **U.S. federal holidays** (travel-behavior context)

Core preparation steps

- wide-to-long reshaping of hourly traffic columns,
- timestamp cleaning/alignment to hourly grain,
- lane aggregation to roadway totals,
- temporal join of traffic + weather + holiday features,
- quality checks for nulls and coverage.

4) Feature Engineering + Modeling Approach

Features we engineered

- Time: hour, month, weekend/holiday flags, peak-hour indicator
- Cyclical encodings: sin/cos for periodicity
- Weather interaction: temperature-dewpoint spread
- Holiday proximity signal
- Lagged traffic features: 1, 2, 3, 6, 12, 24, 168 hours

Modeling path

- Dummy baseline (early benchmark)
- Weekly naive sequence baseline
- Random Forest (mid-project champion)
- **LSTM sequence model (final champion)**

5) Results: Baseline vs Champion

Early benchmark (DummyRegressor)

- RMSE: 295.82
- MAE: 253.32
- R^2 : ~0.00

Final sequence baseline (weekly naive)

- RMSE: 110.57
- MAE: 61.94
- R^2 : 0.8946

Final champion (LSTM)

- RMSE: 106.58
- MAE: 71.40
- R^2 : 0.9021

Takeaway: LSTM improved fit/variance capture and weighted error for multi-step forecasting.

6) The Pivot: What User Testing Changed

Surprising finding

Users trusted the concept, but many were confused by the workflow and model terminology.

Key pain points from Sprint 5

- dashboard felt crowded,
- Random Forest vs LSTM labels were unclear,
- users were unsure what action actually triggers predictions,
- content density + contrast issues reduced readability.

Product changes we prioritized

- onboarding-first guidance,
- simpler prediction path (date/time + clear predict action),
- reduced emphasis on technical model labels,
- improved hierarchy, readability, and control visibility.

7) Live Demo Plan (3-4 Minutes)

Demo scenario

"User planning Saturday morning powder-day traffic"

Walkthrough script

1. Open dashboard homepage.
2. Select realistic forecast date/time window.
3. Trigger prediction.
4. Interpret congestion signal + related context cards.

Success criterion

A user can understand and act on results in under 1 minute.

8) Business Recommendation

Based on model outputs and user-tested workflow, Snowbasin could:

1. **Pre-stage staffing** for predicted high-demand windows.
2. **Send dynamic arrival messaging** to flatten surge behavior.
3. **Use threshold playbooks** (medium/high/severe congestion).
4. **Keep human-in-the-loop overrides** for storms/incidents.

This project is decision support, not autopilot.

9) Extra Credit Deployment Status

Deployment challenge target

Public URL access with no local Python install required.

Current live endpoint

<https://dashboard-snowbasin-wildcats.vercel.app/>

Portfolio impact

A functioning live app strengthens this as a professional artifact beyond the course.

10) Risks, Limits, and Future Work

Known limits

- forecast misses can occur during atypical events,
- MAE tradeoff remains vs naive baseline,
- external incident/event covariates are limited.

Next 3 months roadmap

- event-aware covariates + ensemble residual correction,
- uncertainty intervals + drift monitoring,
- role-based UX + alerting hooks (Slack/SMS/email),
- post-incident forecast-vs-actual review workflow.

11) Closing

Final Message

We turned a semester-long ML build into a user-ready congestion planning tool.

Ask

We welcome feedback on:

- deployment hardening,
- operational threshold tuning,
- feature expansion for winter event conditions.

Thank you - Q&A